

CMPT 230
Group 10 Final Report
-
Slimy Ascent

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Introduction: Game Concept and short analysis - James

A. Overview

- Slimy Ascent is a single-player 2D fast-paced platformer, made using Unity, that follows a weak green slime on his pursuit of strength. Slimy Ascent entails an enjoyable movement-based platforming game with many One-hit knockout environmental obstacles suited for young adults interested in 2D platformers. Slimy Ascent follows an engaging story of the level one weak slime's journey to grow stronger to survive against all that hunt it down while traversing through different environments in a fantasy setting.

B. Key Features

- One-Hit Knockout
 - The environmental obstacles that the player has to avoid as they progress through the game cause the player to die or restart the level whenever they are hit once.
- Movement Based Gameplay
 - The core gameplay is all in the fast-paced movement of the slime through the levels. Compared to some 2D platformers, Slimy Ascent doesn't have attacks and the slime starts at full power. The movement and traversing through the level while avoiding traps are the primary challenges that the player has to overcome.
- Slime as the Playable Character
 - The slime which is generally considered the weakest, or weak in RPG games is what the player plays as. This allows the story to develop as this weak slime is trying to escape from hunters and all that want to hunt it down.

C. Market Relevance and Target Audience

- The target audience of Slimy Ascent is ages 18-24. This targets the largest group of gamers. Statistically, this targets a group of people who, where more than 30% of the people in this age range play platformers. This large percentage allows our game to be explored by potentially a large audience. While other groups do have a higher tendency to play platformers just a bit more, there are a lot more young adults who play games due to their spare time. Slimy ascent is intended to play more difficult so often times it can be very engaging to win challenges or overcome obstacles by learning a simple system.

Game Design Document - James

A. Key Elements

- Game Concept
 - Slimy Ascent is a 2D Linear fast-paced platformer about a Level one Green slime whose journey is to grow stronger from his own kin all while escaping hunters, warriors and alike who would hunt it down
- Mechanics
 - The core of the slime's mechanics is complex mechanics like momentum playing a key role in the fast movement of the slime creature using it to avoid obstacles
 - Vertical and horizontal movement in every direction, By jumping, moving left and right and sliding and jumping off walls
 - As the player traverses the levels in the game, they will have to avoid various hazards. The player has no actual power to "kill" anything - they just need to get to the end of the stage without dying.
 - Hitting a hazard will cause the slime to die and thus, restarting the level. One-Hit Knockout
- Story/Setting
 - The slime character starts off in a large plain just outside of a deep wooded forest where low-level enemies like boars, goblins and other slime creatures roam around aimlessly. Being hunted by strong and weak adventures alike, all the while doing some hunting themselves.
 - The story follows the slime on his escape from the hunters and his pursuit of strength along the way
- Level Mechanics
 - Hazards include:
 - Spikes
 - Pools of water (slimes can't swim!)
 - Potential Hazards that did not make it into the game:
 - Buzz saws
 - Bottomless pits

B. Major Decisions

- The main decision made during the design phase was building the character and setting for the game and then building the stage and hazards around that setting. We initially started with a weak slime that is on the run from hunters, and then turned that in into the different levels in different areas with the different traps that the slime has to avoid.
- The greatest change from the Game Design Document that we implemented into our game was the fairy, Gui. We initially did not have plans for a fairy or a helper but decided that we needed a character that would be on the slime's side and

help him. We decided on a fairy/sprite and eventually made Gui, who helps the slime through the tutorial and other aspects of the game that help the slime progress.

- We initially had thoughts about adding enemies that would impede the slime, other than just the traps but ended up not implementing that into the game, due to time constraints and a heavier focus on the trap hazards themselves.

Character Design - Zhire

A. The Slime

a. Visual Design

- The slime is small, green, and has beady eyes and rosy cheeks. It is semi-transparent so you can see through it to the other side.
- The slime bounces in place as it stands idle to show some liveliness. When the slime moves, it shifts itself in the direction that it's going.

b. Personality Traits

- The slime is timid and shy but is mostly desperate, as other adventurers are constantly hunting it down. This is reflected in the game by its agile movement in an attempt to avoid hazards and obstacles that come its way.
- The slime has no mouth, so it has no way to speak. To communicate, it wriggles its body, jumps, or shakes itself.

c. Narrative Role

- The slime is a stereotypical weak creature in fantasy worlds; therefore, it lives in constant fear, trying to escape its captors, which we sadly didn't have the time to implement.
- The slime is a stereotypical weak creature in fantasy worlds; therefore, it lives in constant fear to survive. Along with Gui, its fairy companion, their main goal is to escape from other adventurers, which we sadly didn't have the time to implement.

d. Visual Animations



Idle



Moving



Jumping



Wall Sliding

e. Design Process

- The slime is based on slimes that you would see in a medieval fantasy game. Green slimes are also the standard color of slimes, so I went with that. I made it so it had big eyes and rosy cheeks to convey kindness and innocence.

a. Visual Design

B. The Fairy

- a.** The fairy, also known as Gui, is a small creature wearing a blue dress with butterfly-like wings with a similar shade of blue.



Gui

b. Personality Traits

- The fairy is a kind-hearted soul and always talks in a happy-go-lucky way. It's naturally helpful towards other weaker creatures, such as the slime.

c. Narrative Roles

- The fairy's main role in the game is to help the player progress through the game. At the start, she provides them with visual inputs to let the player know the controls for the game.

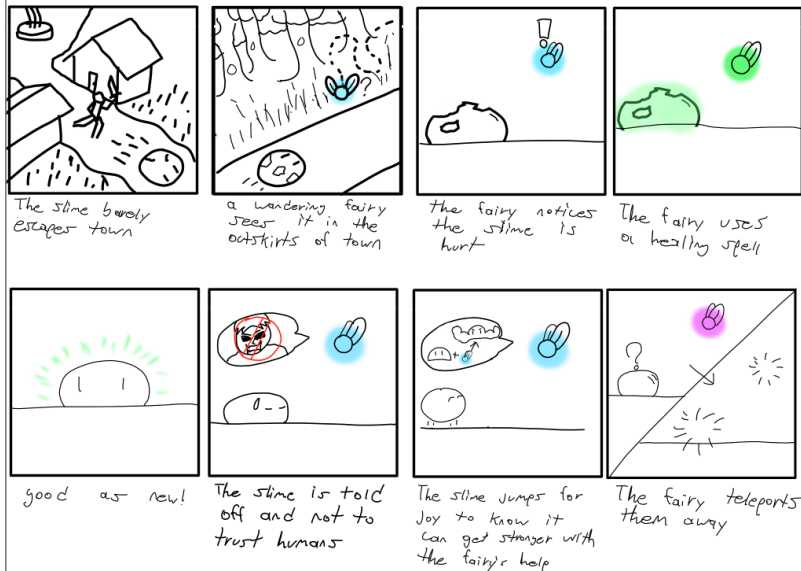
d. Design Process

- The character's design is inspired by fairy godmothers portrayed in books and movies. I chose to go with a blue color palette to portray her goodness of heart.

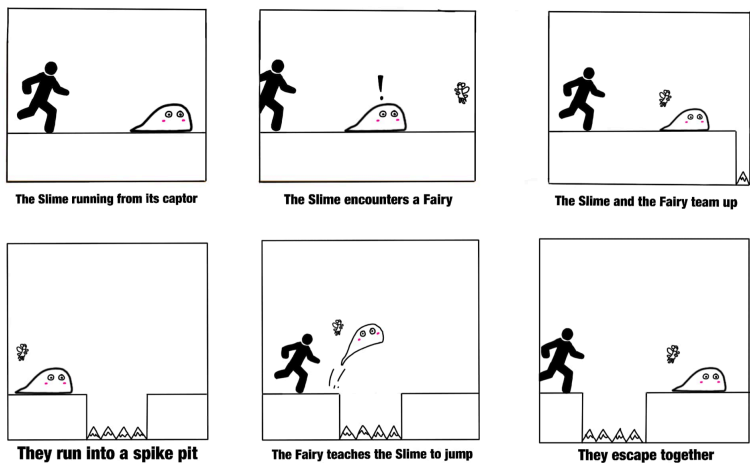
Storyboarding and Dialogue

A. Storyboards

a. Eli



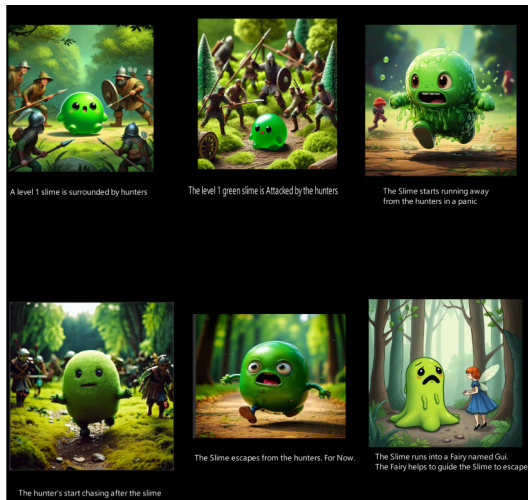
b. Zhire



Slimy Ascent Storyboard

Zhire

c. James



d. Stefan



B. Dialogue

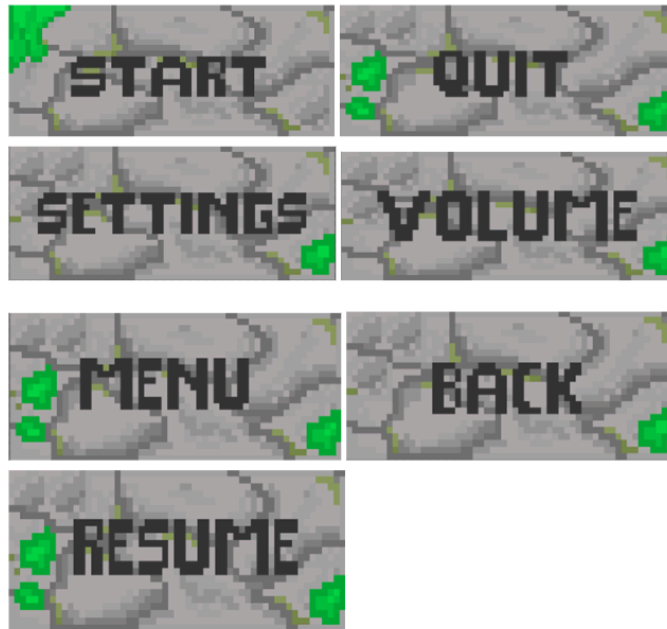
- For our player character, they lacked any audible dialogue as they are a mute - only communicating through wiggles and slight movements. Dialogue primarily plays a part through our tutorial character, Gui; As the player begins the game, Gui will have dialogue boxes pop up as the player progresses through the tutorial level - with the dialogue changing depending on what the player needs to do next.

UI Design - Eli and Zhire (Assignment 5)

- The title text was created in Krita and Piskel by Eli. To make it feel more immersive slimy is a bouncy green but almost liquid texture to imitate the look of the slime as well to have the main character above the slime text. The brick-like structure of ascent was supposed to relate to the tower that isn't properly shown in the prototype
- The UI buttons were mainly created in Piskel by Zhire. We went for a medieval fantasy theme because we believe it fits our game the best. I kept it simple by drawing out the buttons in a mossy rock texture while also giving players a sense of exploration and

adventure while playing the game. I also added bits of slime on the buttons to reference the playable character.

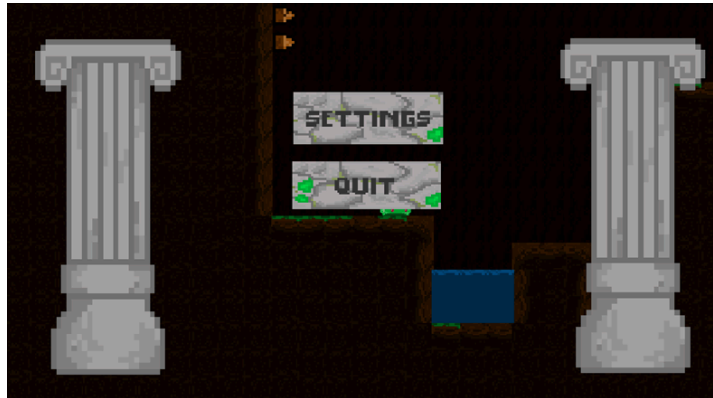
a. UI Buttons



b. Main menu



c. Pause menu



Sound Effects and Music - Eli

Slimy ascent's music comes from incompetech; a site for royalty free music. All where made by "Kevin MacLeod" I gathered 3 songs from his website:

- *"Voxel revolution"*

A song for the title screen, it fit a explorative feel and felt expressive enough to hook players in

- *"Cyborg ninja"*

Despite its name this song was used for the cave section of the game, it had a bit more energy to it making it feel like you need to move around and progress through the level

- *"Cruising for goblins"*

Another bouncy feeling song, it fit very nicely as it's upbeat and still fast paced.

For SFX I created some using help from presets from jsfxr. The only 2 present in the game are the death sound and the jump sound. The Death sound is a simple thud letting the player know they died. The jump sound imitates what a living slime's jump could feel like within the context.

Prototype Showcase

[Prototype Showcase](#)

Challenges and Learning

A. Challenges:

- Unity was exciting to learn but difficult to grasp at the start, with its many layouts and very many buttons all with sometimes new words to comprehend.
- Sound was too difficult to work with as none of us had any experience at all in how it works. However it worked in our favor to have more work on everything else and use free assets for music
- Zhire: Making graphical assets was a challenge as I didn't have any experience with pixel art. I had to ask my teammates every time I made an asset to ensure that it looked good and fit the theme we were going for.
- Stefan: Finding a movement system/script that matched the feeling of platformer we wanted. As games like Super Meat Boy were a large inspiration for our game, we wanted a heightened focus on control over your character, but also for the movement of the character to be "exemplary" of the player character themselves i.e. A bouncy, mobile slime should have a relatively bouncy feeling to it - one that we believe we managed to achieve with the movement system we implemented in the game. (NOTE: I followed a tutorial/youtube video in order to implement this movement script; the actual math behind it was quite complicated).
- Version control initially was an issue, and we began with sharing our assets/project files through google drive; however, we relatively quickly settled with GitHub - ensuring to follow the correct procedures so files wouldn't go missing (which seemed to be a common issue).

B. Learning:

- Unity is now a skill we all can safely learn without struggle just by getting the basics down.
- Sufficient level of communication between group members. Often times we would have an issue arise for a particular member (i.e. issues importing assets for a tilemap) and we would have to work together and figure out how to solve the problem before either could continue working. The aforementioned is simply an example, but a strong level of communication really helped us ensure everyone was on the same page.
- Ability to find sources/assets online and reformat/restructure to work within the context of what we're working on.
- Creating assets on our own (primarily visual/art), despite not having prior experience of doing so.

Personal Reflection

Eli Born: Making slimy ascent was pretty fun and was honestly faster than I thought it would be. There would be quite a bit of quality of life to add from my perspective and a few issues that I noticed that wouldn't really matter in the prototype. I will definitely be heading into the next gaming class with a better understanding

James Dwyer: I enjoyed making Slimy Ascent and, in general, game creation. It taught me a lot about unity itself and how to implement various features in a game. It also allowed me to gain a better understanding of using GitHub in projects, as well as cooperating and communicating with other people on a group project. One of the challenges I faced was learning Unity, as well as trying to find a role and contribute towards the project at the start of the project. If I could redo the project I would be more involved in the beginning stages and clearly communicate with my group members.

Zhire Pelayo: Making Slimy Ascent was a fun experience. Not only that, but it taught me a bunch of new skills, mainly in Unity and GitHub, that I'm sure will be useful in the future. If I could give a tip to myself at the beginning of the semester, I would tell him to watch more tutorials instead of trying to figure things out himself.

Stefan Cvijovic: I thoroughly enjoyed the development process of Slimy Ascent; as an avid fan of 2D platformers, it was enlightening to be actively involved in the process of the actual physics and movement of the game itself - as to me, that's what is most important in such a game. The process tested and reinforced my ability to coordinate with my teammates, as well as taught me how to learn on the fly when it came to topics/concepts I had little experience with. Looking back, if I had to change how I approached the development of our game, I would have tried to allocate more time early on to testing the prototype, as well as trying my hand at writing and developing my own movement script. The latter is something I will definitely be considering moving forward into CMPT 330.

References

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https://www.youtube.com/watch?v=KbtcEVCm7bw&ab_channel=Dawnosaur Tutorial for script
<https://github.com/DawnosaurDev/platformer-movement> - Codebase for script
<https://www.youtube.com/@GameCodeLibrary/videos> - General tutorials followed for platformers